

Deferred readouts in the two-stage bit interchange

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This note describes the round-seven bit side. It keeps the round-six interchange (flag basis, retained point totals with copied centres, side roles batched one level down, data-entrance run) and changes three things: the side program comes from PR #62's producer, its frames are lifted and its fan-out copies are made late, and the stage-one word is reordered so that most slots read their garbage out at a nonzero frame. All objects are frozen in `certificates/round7/`; `scripts/certificate_round7.py` rebuilds the child-width histogram from the schedule, and the checks in `independent/deferred-readout/` verify every fact used below. Throughout, $h = 23$, $m = h^2$, $v = \binom{h}{3}$, $N = v^2$, $G = I - J/9$, and for a triple T the root frame of the output wire y_T is $t_T^\perp = \ker(9t_T - 31)^\top$; the frame of a retained total of c is $U_c = \ker(\mathbf{1} - 3e_c)^\top$.

1 The side program: PR #62's producer

The side DAG has the v triples as leaves and disjoint additions as nodes; its roots are the $3v$ outputs $\text{out}(c, T) = \{S : S \cap T = \{c\}\}$ and the h retained totals $\text{star}(c)$. We build it with PR #62's producer (ikeboy): interval strips and core-aware pair assembly, with PR #41's alternating order and links (RaD/hipotures), and links chosen by a moment-weighted matching in the spirit of PR #44 (rohanarun). A link hands the non-pivot slot of a donor gate to a later use of the same value, so the number of slots is $R = \#\text{additions} + 3v + h - \#\text{links} = 28866$. Only the DAG enters the certificate, and it is checked from the definitions: every addition is disjoint, every output and total has the support above, and every node is reachable from a root (`check_lifted.py`, part A; `check_word.py`, part A).

2 Lifted frames and late copies

For an addition n let U_n be the common kernel of the root covectors reachable from n and F_j the span of the first j rows of a fixed integer flag matrix. The gate of n acts at

$$M_n = \text{span}(n) + (U_n \cap F_{j(n)}),$$

with a lift index $j(n)$ per addition. This is Paurel's complement-frame construction: it is valid for any frames with (H1) leaf frames $\langle t_S \rangle$, (H2) frames nested along every slot (DAG edges, links, copies), (H3) roots inside $t_T^\perp$ or U_c , (H4) G nondegenerate on every frame. Nesting is a theorem given $j(n) \leq j(t)$ on every slot successor pair, $\text{span}(x) \leq \text{span}(t)$ on every link and $\text{span}(n)$ inside every root it reaches directly; then $U_n \leq U_t$ and $M_n \leq M_t$. In this witness every lifted frame equals U_n . Dimensions are exact (modular law, ranks with Hadamard certificates or Bareiss).

Late copies. A node with several free uses is copied for each of them. A copy made early sits at M_n ; a *late* copy for the user ℓ_i of an ordered list $\ell_1, \dots, \ell_k$ is made when the pivot has climbed to

$$C_i = \bigcap_{i' \geq i} M(\ell_{i'}), \quad M_n \leq C_1 \leq \dots \leq C_{k-1} \leq M(\ell_k),$$

and starts there, so the copy saves the rank $\dim C_i - \dim M_n$. Each copy is placed as late as possible (before its user's gate, the pivot's next gate and the next copy). The intersections are needed: without them consecutive pivot frames are not nested (1396 of 1396), and a copy moved after the pivot's next gate breaks the replay.

3 The B-defer word

In stage one each slot u starts with its garbage z_u , and the word reads it out of every target it reaches: $y_T \leftarrow C_{T,u} z_u$, where $C = JL$ is given by the transpose sweep of L . In rounds five and six all these readouts happen at frame 0, before the second pass. B-defer splits the second pass in two phases.

- *Phase 1* is the closure of the retained totals' last operations under per-slot sequence precedence: every op that must precede a total's completion. The totals complete, and their copies read every target containing their centre at 0 (copied centres, as in round six).

- A slot that phase 1 neither reads nor writes still holds its garbage after phase 1. It reads it out *now*, at a frame $\sigma_u \leq F_0(u) \cap \bigcap_T t_T^\perp$ (over its targets T), then the rest of L runs, then the output reads at $t_T^\perp$, then L^{-1} and V^{-1} exactly mirror it.

Only the topological order of the word changes, so it is still a mirror and still restores arbitrary scratch. The slot's chain now starts at σ_u : its corner has rank $r_u = h - \dim \sigma_u$ and its auxiliary edge rank $m - r_u$.

The sorting rule. On a target y_T the readout frames, in time order, are 0 (non-deferred slots and centre copies), the deferred σ_u , and $t_T^\perp$; they must form a chain. The word reads the deferred slots in increasing $\dim \sigma_u$, and applies the deferred V gates (below) in increasing $\dim F_0$. With this rule every y_T sequence is nested in time order, exactly over $\mathbb{Q}$, for both the $\mathbb{Z}$ and the $\mathbb{F}_2$ supports of C ; the reversed order breaks 11879 chains.

4 V leaves as late copies on X_S

A leaf S is used by several gates. Instead of copying x_S by fan-out, every use i gets its own V gate on the data wire X_S , into a slot that starts at

$$s_i = N_i \cap N_{i+1} \cap \cdots \quad (\text{falling back to } \langle t_S \rangle \text{ if } G \text{ is degenerate on it}),$$

where N_j is the first frame of use j . The V gates are then late copies on X_S : its frame chain $\langle t_S \rangle \leq s_{i_1} \leq s_{i_2} \leq \cdots \leq F$ has total rank $h - 1$, as before, and the use slots start at s_i instead of $\langle t_S \rangle$. The V gates of deferred slots come after their readouts; in time the X_S chain is $\langle t_S \rangle$, the early V frames (by dimension), the deferred V frames (by the rule above), F , and it is checked nested exactly in that order. The number of slots is unchanged.

5 What the certificate uses

Only the $\mathbb{F}_2$ flag identity. The bit network is used over $\mathbb{F}_2$: the word must add exactly x_T to every y_T and restore every slot, for arbitrary scratch. The replay checks this over $\mathbb{F}_2$ with the parity coefficients (the actual circuit) and, separately, over $\mathbb{Z}$ with the exact path-count coefficients; the histogram and a_b are the same with either support.

The stage-two complement. Stage two is the complement time-reversal of stage one, with the same edge multiset. This is the same assumption as in round six; it is stated, not machine-checked here.

Frames and genericity. Every σ_u and s_i is a primitive integer basis (entries at most 13). Exactly over $\mathbb{Q}$: $\sigma_u \leq F_0(u)$, $\sigma_u \leq t_T^\perp$, $t_S \leq s_i \leq N_i$, G nondegenerate. Every high-rank step ($2r > h$) that the reordered word adds (slot starts, gauged corners $\sigma_u \rightarrow F$, y_T and X_S chain steps) has the generic north-east pivots under both conjugations at the integer point of `check_lifted.py`, so the side lemma batches it.

6 Accounting

Per stage and invocation: every slot contributes its auxiliary edge (block $m - 2r_u$) and its corner, both batched by the inner lemma, and its chain steps; every centre copy its step $U_c \rightarrow 0$ of rank $h - 1$; every y_T and every X_S its chain of total rank $h - 1$; the data entrance its block $m - 4h + 2$ and corner; one copy-correction singleton per pair. The rank sum is exactly $s = Wm - N + L$ with $W = 2N + 2vR$, $L = 2vh(h - 1)$, so the histogram re-partitions the round-six budget. The moment certificate gives

$$a_b = 31987/(5 \cdot 10^8) \quad (\text{staircase entrance corner, runs } h - 2, h - 5, 1^6),$$

$$a_b = 12599/(2 \cdot 10^8) \quad (\text{round-six entrance corner}),$$

and with round six's complex interchange ($a_c = 36926111/(5 \cdot 10^{11})$, $h = 24$) the assembly gives $\kappa = 1599247689723/(2.5 \cdot 10^{16}) \approx 6.39699 \times 10^{-5}$ with the staircase corner and $\kappa = 6299103187973/10^{17}$ with the round-six corner. The staircase corner is certified by `check_stair.py` at the same integer point; the entrance edge is not changed by the reordering.